COMPLETED MINI-PROJECT · 04 NOVEMBER 2026

Uncertainty and sensitivity of flow in a 2D random network

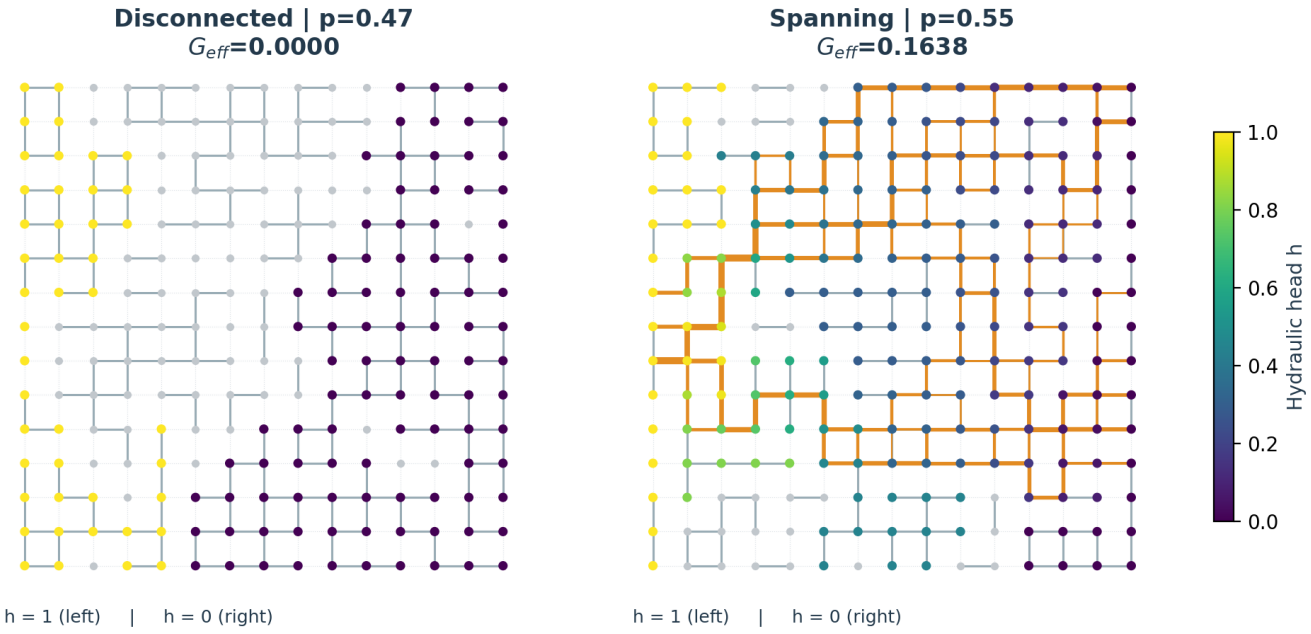
> **Finding.** Connectivity controls both the existence and magnitude of flow. Broadening the uncertain connectivity distribution increases mean conductance strongly, while the probability of a spanning path stays close to one half. Conductance heterogeneity changes flow magnitude, with bottlenecks dominating near the connectivity threshold.

At σ=1, increasing δ from 0 to 0.25 changes mean effective conductance from **0.0183** to **0.0992** (5.44×). In the latter case, 48.62% of realizations carry no flow. These results come from 248,188 solver calls, including 90,000 main uncertainty realizations.

Model and scope

A square bond lattice has n=25 nodes per side (625 nodes; 1,200 potential links). All quantities are dimensionless. Each realization samples one global p; links are independent conditional on that p. The left/right hydraulic heads are 1/0; the top/bottom boundaries have no exterior links and hence no imposed flux.

$$p = 0.5 + \delta U,\ U \sim \mathrm{Uniform}(-1,1)$$
$$g_e = \mathbb{1}[W_e < p]\exp(\sigma Z_e),\ W_e \sim \mathrm{Uniform}(0,1),\ Z_e \sim \mathrm{Normal}(0,1)$$
$$Q_{ij} = g_{ij}(h_i - h_j),\ \Sigma_j Q_{ij} = 0,\ G_{eff} = Q_{total} / \Delta h = Q_{total}$$



Illustrative 15×15 networks. Node colors show head; orange link widths show flow magnitude. Grey nodes are floating components with undefined absolute head.

1 · Numerical method and verification

Build the weighted graph Laplacian L. Connected-component analysis identifies clusters touching both fixed-head boundaries. Solve only free nodes in these clusters using SciPy's sparse direct solver:

$$L_{FF} h_F = -L_{FB} h_B$$

Clusters touching only one boundary have constant head and zero flow. Interior floating clusters have arbitrary absolute head (stored as NaN) and zero flow; they are excluded from the singular linear system. Total inlet flux is evaluated on spanning clusters. This avoids artificial leakage or regularization.

Independent check	Expected result / acceptance
Full uniform grid, n=2,3,9,25; g=0.1,1,7	G = g n/(n−1), h linear
Single series path / two parallel paths	1/Σ(1/g) / sum of branch conductances
Disconnected islands / vertical cut	G = 0; floating heads undetermined
Conservation / energy / head bounds	Qin = Qout = Σ g(Δh)²; 0 ≤ h ≤ 1
Scaling / Rayleigh monotonicity	G(cg)=cG(g); increasing a link cannot reduce G
Sensitivity estimator benchmark	Ishigami analytic indices, absolute tolerance 0.018

Verification: all 24 automated tests pass. For 1,000 additional random networks, maximum interior balance residual was 2.75e-13, maximum relative inlet/outlet mismatch 7.91e-12, and maximum energy discrepancy 2.41e-13.

Experimental design and efficiency

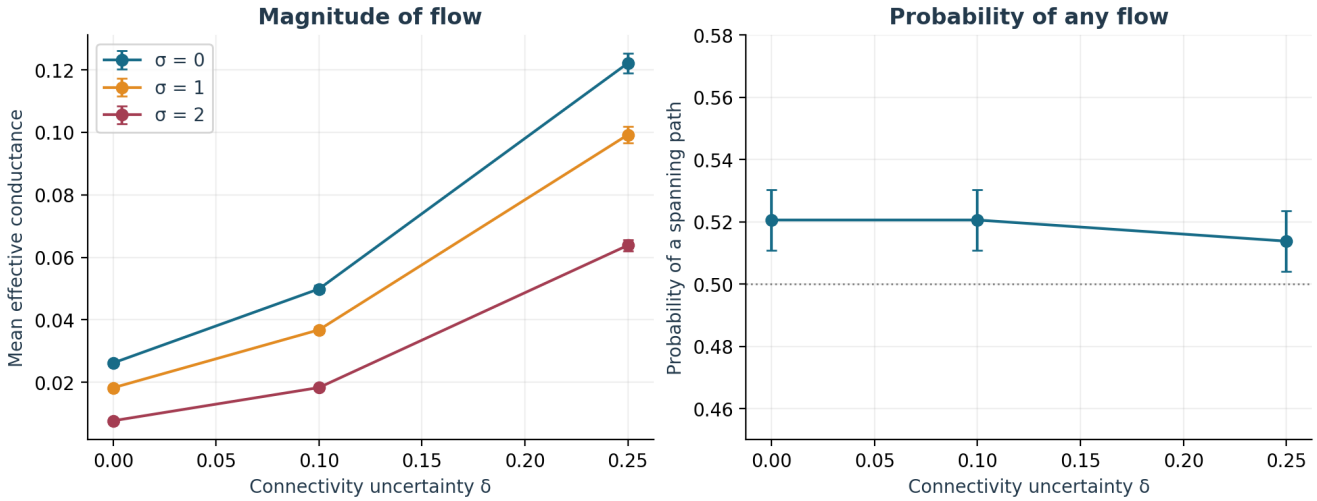
Experiment	Design	Solver calls
Uncertainty	3 δ × 3 σ × 10,000	90,000
Connectivity sweep	p=0.30:0.05:0.70; 2,500 each	22,500
Additional finite sizes	n=15,41; δ=0,0.25; σ=1; 5,000 each	20,000
Grouped Sobol	16,384 pairs × (5+2)	114,688
Solver diagnostics	1,000 independent networks	1,000
Total	n=25 size comparison reuses scenario data	248,188

n=25 resolves the connectivity transition at low cost; n=15 and 41 check size dependence. δ∈{0,0.1,0.25} spans p=0.5 through p∈[0.25,0.75]. σ∈{0,1,2} spans constant conductance to strong heterogeneity (σ²=0,1,4). σ, rather than σ², is sampled uniformly for sensitivity. Seed 20261104; separated RNG streams; common random numbers across scenarios; measured full study time 188.1 s on this execution host. Hardware and timing are not a performance guarantee.

2 · Uncertainty: statistics of predicted flow

Each row contains 10,000 independent network realizations. The same latent realizations couple different rows to improve paired comparisons; rows must not be pooled as independent observations. P(G=0) is a probability, distinct from the size of nonzero flow.

δ	σ	Mean [MC 95% CI]	P(G=0)	95th pct.	Mean, G>0
0.00	0	0.0262 [0.0256, 0.0268]	47.94%	0.0805	0.0503
0.00	1	0.0183 [0.0178, 0.0187]	47.94%	0.0590	0.0351
0.00	2	0.0077 [0.0075, 0.0079]	47.94%	0.0288	0.0148
0.10	0	0.0499 [0.0487, 0.0511]	47.94%	0.1775	0.0959
0.10	1	0.0368 [0.0359, 0.0377]	47.94%	0.1360	0.0707
0.10	2	0.0183 [0.0178, 0.0188]	47.94%	0.0752	0.0352
0.25	0	0.1222 [0.1190, 0.1253]	48.62%	0.4491	0.2378
0.25	1	0.0992 [0.0966, 0.1019]	48.62%	0.3838	0.1931
0.25	2	0.0639 [0.0620, 0.0657]	48.62%	0.2760	0.1243

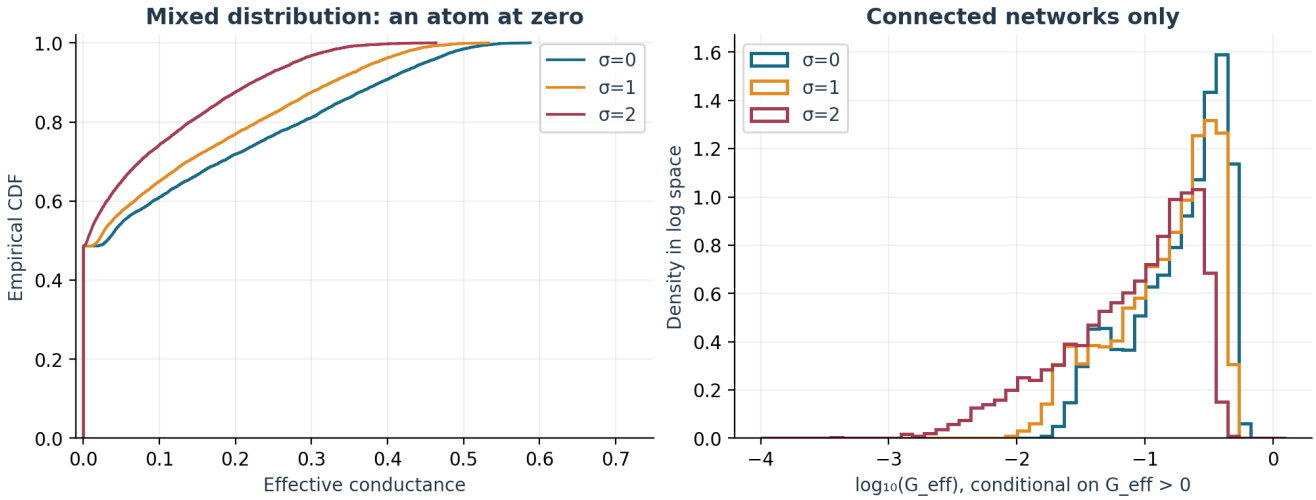


Mean conductance: normal Monte Carlo 95% intervals. Spanning probability: Wilson 95% intervals; identical across σ by construction.

Reference scenario (δ=0.25, σ=1). Mean = 0.0992 [0.0966, 0.1019]; standard deviation = 0.1341; median = 0.0195; 95th percentile = 0.3838 (order-statistic 95% interval [0.3778, 0.3897]). The probability of a spanning path is 0.5138, with Wilson interval [0.5040, 0.5236]. Conditional mean given a positive flow is 0.1931.

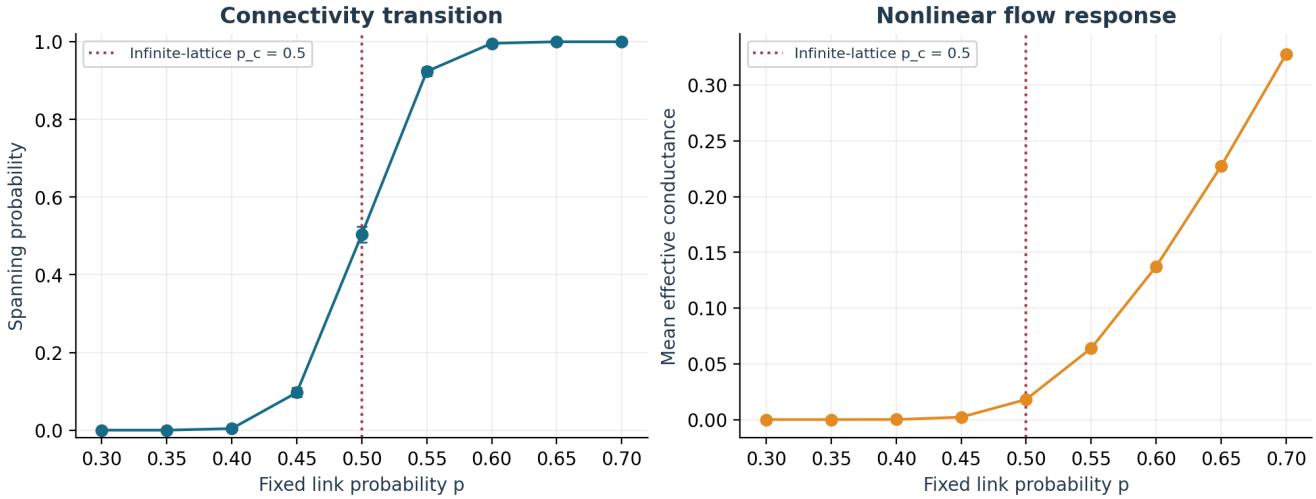
The reported confidence intervals quantify finite Monte Carlo sampling error under the specified model. They are not prediction intervals for a new network and do not include structural model uncertainty. Quantile intervals use binomial order statistics and may be conservative when observations tie at zero.

3 · Distribution and connectivity transition



The full output law has a discrete mass at zero and a broad positive component. A single lognormal distribution cannot describe the full output.

For the reference scenario, the atom at zero is 0.4862; the mean, median and upper tail are far apart. Report P(G=0) and positive-flow statistics separately. The density plot describes $\log_{10}(G)$ conditional on positive flow, not the density of G. The CDF panel focuses on $0 \leq G \leq 0.75$; full values are retained in the raw data.



Fixed-p sweep, n=25 and σ=1, 2,500 samples per p. p_c=0.5 is the infinite square-lattice bond threshold [2]; the finite network has a smooth crossover.

Increasing δ leaves E[p]=0.5, but spreads realizations below and above the connectivity transition. The nonlinear flow response gives high-p networks substantial flow, while low-p networks remain at zero. This explains the increased mean without a comparable increase in spanning probability; it is an empirical result over these ranges, not a general convexity theorem.

Unconditional links are correlated because they share p: for different edge indicators $I_e, I_{f'}$, $\mathrm{Cov}(I_e, I_{f'}) = \mathrm{Var}(p) = \delta^2/3$. Sampling a separate p for every edge would remove this mechanism and implement a different model.

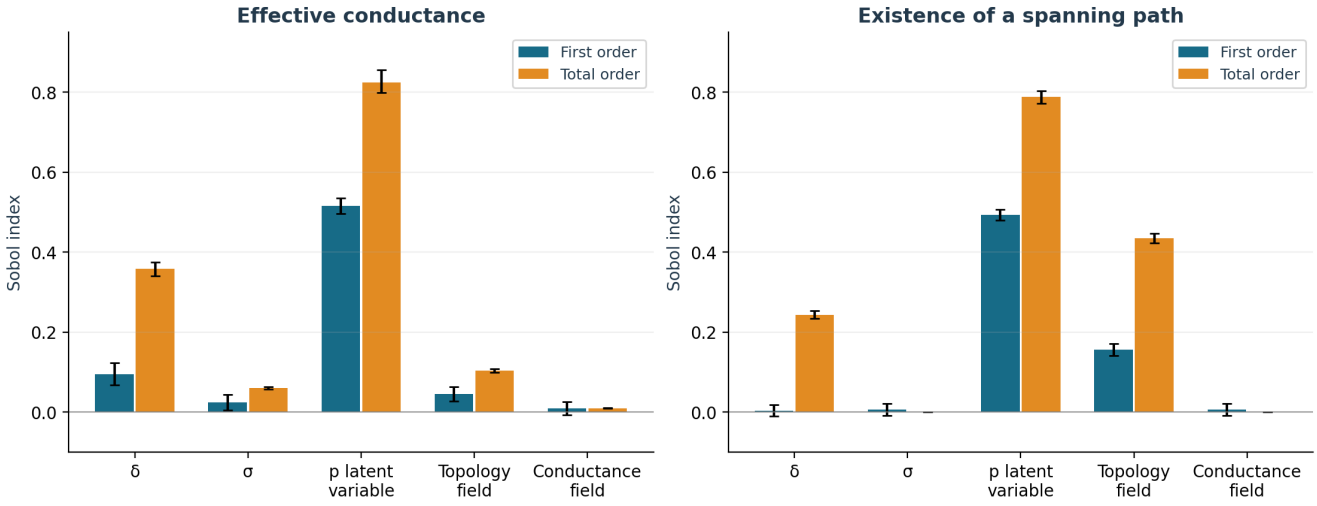
4 · Global sensitivity with stochastic inputs

Use grouped variance-based Sobol analysis [1], which captures threshold behavior and interactions. Expand the stochastic simulator into five independent input groups: δ~U(0,0.25), σ~U(0,2), U~U(−1,1), topology field W (1,200 uniforms), and conductance field Z (1,200 normals). This avoids incorrectly treating δ and the realized p as independent inputs.

$$S_i = \mathrm{Var}(E[Y \mid X_i]) / \mathrm{Var}(Y)$$
$$S^T_i = E[\mathrm{Var}(Y \mid X_{-i})] / \mathrm{Var}(Y)$$
$$\hat{S}_i = 1 - \mathrm{mean}[(Y_B - Y_{AB_i})^2]/(2V),\ \hat{S}^T_i = \mathrm{mean}[(Y_A - Y_{AB_i})^2]/(2V)$$

A and B are independent grouped input samples; AB_i replaces group i of A with B. V is the pooled sample variance of Y_A and Y_B. Use 16,384 independent base pairs (114,688 evaluations), with 1,000 paired-row bootstrap resamples for percentile 95% intervals.

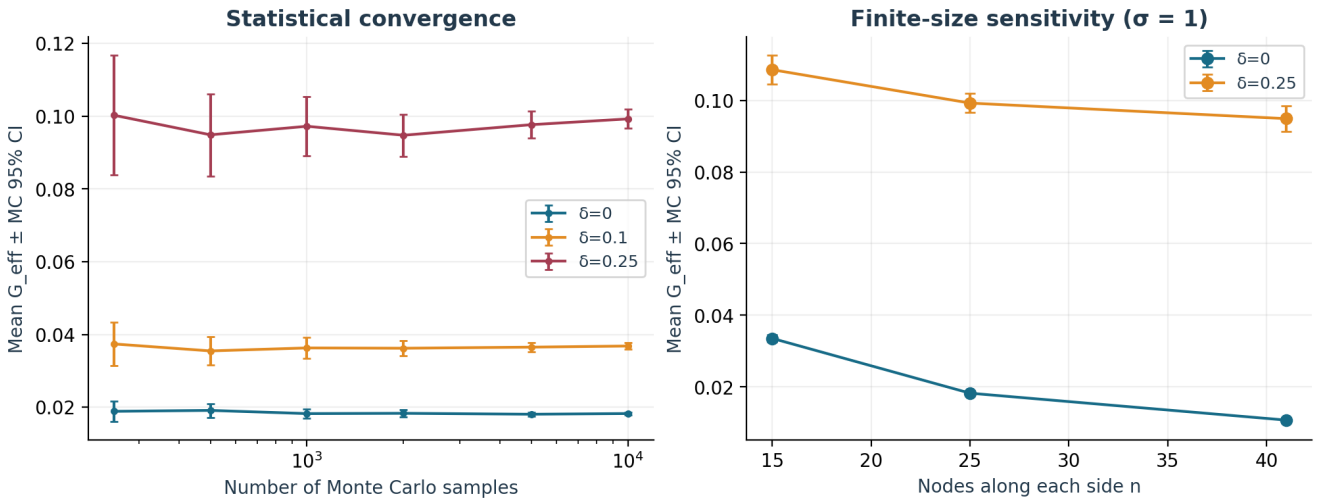
Group	First order [95% CI]	Total order [95% CI]
δ: uncertainty width	0.096 [0.068, 0.123]	0.358 [0.341, 0.375]
σ: log-conductance spread	0.025 [0.005, 0.044]	0.061 [0.058, 0.064]
U: global connectivity draw	0.516 [0.496, 0.536]	0.825 [0.799, 0.856]
W: topology field	0.046 [0.028, 0.063]	0.104 [0.100, 0.109]
Z: conductance field	0.010 [-0.007, 0.026]	0.010 [0.010, 0.011]



First-order and total-order indices for flow magnitude and the binary existence of a spanning path. Error bars are bootstrap 95% intervals.

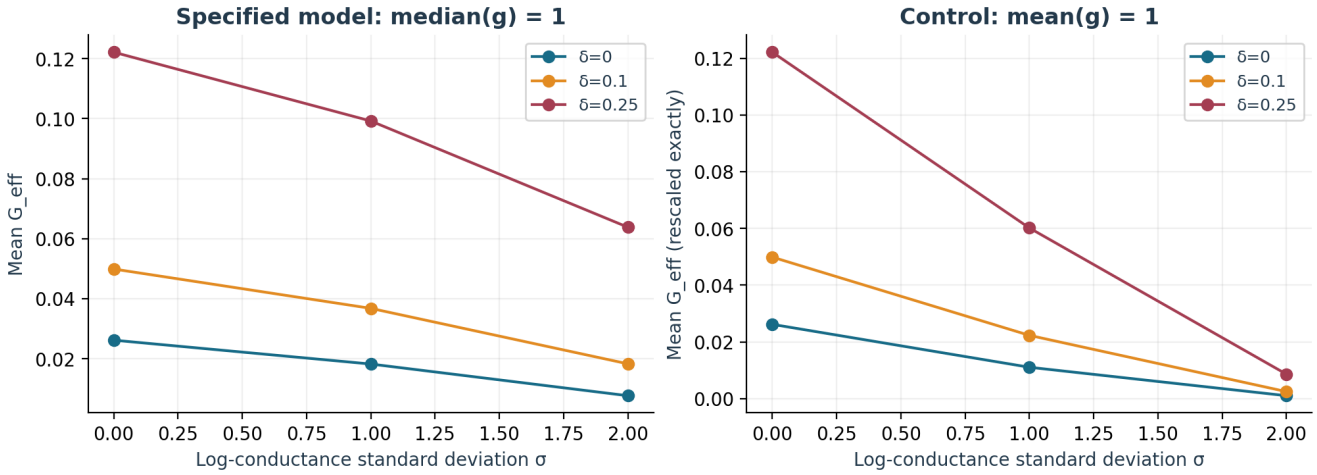
U dominates; δ has a much larger total than first-order effect, indicating interactions, principally with the connectivity draw. About 30.7% of variance is assigned to interactions by the sum of estimated first-order indices. Total indices overlap and must not be added as independent percentages. σ and Z have exactly zero total effect on spanning because every active conductance is positive. Small spurious first-order estimates for known inactive factors are sampling error; raw estimates are preserved. Rankings depend on the chosen priors and latent parameterization and concern all realizations, including zero-flow networks.

5 · Convergence, finite size and normalization



Left: nested Monte Carlo checkpoints (250 to 10,000). Right: different network ensembles at n=15,25,41; this is finite-size sensitivity, not continuum mesh convergence.

The final relative 95% Monte Carlo half-width of the mean is 2.19–2.91% across the nine scenarios. At p=0.5 (δ=0, σ=1), mean G decreases from 0.0335 at n=15 to 0.0183 at n=25 and 0.0107 at n=41. Finite-size effects remain material near the threshold. Results cannot be advertised as a size-independent continuum prediction. Sobol prefix convergence is saved in figure 08 and CSV; the leading ranking is resolved, while tiny first-order terms remain uncertain.



The assigned model fixes median(g)=1, so increasing σ also increases E[g]. A constant-mean control is obtained by exact per-network rescaling.

In the specified model E[g | active]=exp(σ²/2), equal to 1, 1.6487 and 7.3891. Nevertheless mean G falls as σ increases in the studied scenarios: for δ=0.25, from 0.1222 to 0.0639. Weak links on series paths constrain flow. To isolate variability at fixed mean link conductance, set g*=g exp(−σ²/2); linearity gives G*=G exp(−σ²/2) exactly. This is a supplementary control and does not replace the assigned distribution.

6 · Interpretation, limitations and extensions

Main findings

- Connectivity and its interactions dominate global output variance over the declared ranges. Characterizing connectivity has more value here than measuring every individual link conductance precisely.
- The mean alone hides a near-50% probability of no flow and a highly variable positive component. A useful flow forecast includes connection probability, conditional flow and an upper quantile.
- Broader uncertain connectivity increases mean flow at fixed E[p], while greater log-conductance spread reduces mean flow in this study. Both are range- and model-dependent findings.

Limitations

- Regular, isotropic, two-dimensional bond geometry; no fracture length/orientation distributions, intersections, aperture physics or spatially correlated fields conditional on p.
- Steady, single-phase, linear hydraulic conductance; no storage, transport, transient pressure, matrix flow, multiphase processes or geomechanical feedback.
- Ranges are exploratory, not calibrated priors. The model is dimensionless and has not been validated against field data. Analytical and invariant checks establish implementation correctness, not geological predictive accuracy.
- Finite-size dependence is significant. Near-critical paths, rare tails and strongly heterogeneous conductances need more sampling and conditioning checks if ranges are extended.

Extensions to fractured rocks

Generate discrete fracture networks in 2D/3D with measured fracture geometry and intersection statistics; link conductance to aperture and the cubic law with consistent units; introduce anisotropy and spatial correlation; calibrate against heads, flow or tracer data. Include matrix exchange and transient transport, then propagate parameter and structural uncertainty. Surrogates could accelerate these richer models after validating their errors near the connectivity transition; they are unnecessary for this inexpensive lattice study.

Reproducibility and deliverables

Source solver, 24 tests, complete compressed realization-level CSVs, summary CSVs, seed/version manifest, scripts for figures/reports/slides, and an executed notebook are included. The presentation is supplied as editable PowerPoint and PDF. Run commands are in the repository README. All reported values are generated from saved results, not manually populated example data.

References

[1] M. J. W. Jansen (1999). Analysis of variance designs for model output. *Computer Physics Communications* 117, 35–43. doi:10.1016/S0010-4655(98)00154-4.

[2] H. Kesten (1980). The critical probability of bond percolation on the square lattice equals 1/2. *Communications in Mathematical Physics* 74, 41–59. doi:10.1007/BF01197577.

[3] Project brief: *Mini-project: Uncertainty and Sensitivity Analysis of Flow in a 2D Random Network*, supplied PDF, two pages. Parameter ranges, sample counts and numerical implementation were selected for this study.